Verifying Distributed Protocols with Proof Assistants: A Survey of Approaches

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This survey evaluates several approaches for verifying distributed algortihms with automated theorem provers. /L

CCS Concepts: • **Security and privacy** → **Logic and verification**.

Additional Key Words and Phrases: distributed algorithms,

ACM Reference Format:
Maxwell Moir. 2026. Verifying Distributed Protocols with Proof Assistants: A Survey of Approaches. *J. ACM* 37, 4, Article 111 (August 2026), 2 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

ACM's consolidated article template, introduced in 2017, provides a consistent LaTeX style for use across ACM publications, and incorporates accessibility and metadata-extraction functionality necessary for future Digital Library endeavors. Numerous ACM and SIG-specific LaTeX templates have been examined, and their unique features incorporated into this single new template.

If you are new to publishing with ACM, this document is a valuable guide to the process of preparing your work for publication. If you have published with ACM before, this document provides insight and instruction into more recent changes to the article template.

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1.1 Motivations

As noted in the introduction, the "`acmart`" document class can be used to prepare many different kinds of documentation — a double-anonymous initial submission of a full-length technical paper, a two-page SIGGRAPH Emerging Technologies abstract, a "camera-ready" journal article, a SIGCHI Extended Abstract, and more — all by selecting the appropriate *template style* and *template parameters*.

2 PRELIMINARY

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ACM 0004-5411/2026/8-ART111
https://doi.org/XXXXXXX.XXXXXXX

3 METHODOLOGY

3.1 Literature Search Strategy

To identify relevant papers, we used a systematic literature search strategy. We chose to query two databases, the ACM digital library and IEEE Xplore. These databases were accessed on April 17 2026.

To ensure the quality of of this review, we decided to follow the PRISMA framework to inform our search strategy. The search strategy can be summarized into multiple steps. First, we curated relevant keywords and constructed seach queries to compile relevant papers based on the guiding questions of the survey. This led to the following seach query:

("proof assistant" OR "theorem prover" OR "mechanized verification" OR "formal verification") AND ("distributed protocol" OR "distributed algorithm")

This query yeilded 213 results from the ACM Digital Library and 49 results from IEEE Xplore.

3.2 Inclusion and Exclusion Criteria

These results were then screened based on standard inclusion and exclusion requirements. This reduced the result count to 37 for ACM and XX for IEEE Xplore. From this point, the results were expanded by adding commonly cited papers.

4 APPROACHES

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5 FUTURE DIRECTIONS

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6 CONCLUSION

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009